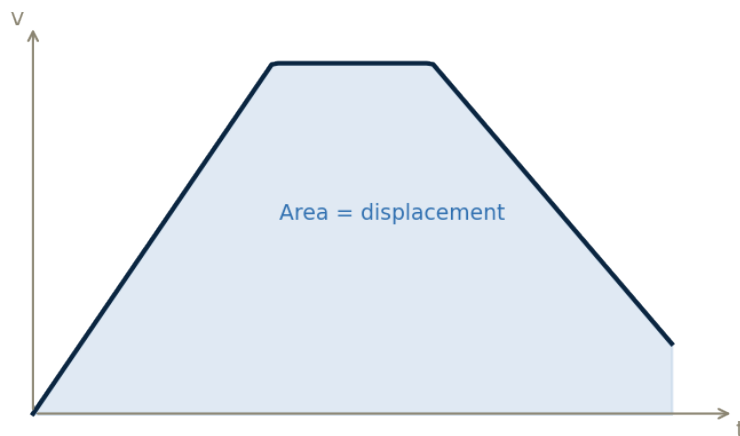


Kinematics

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Displacement is a vector (shortest straight-line path with direction); distance is a scalar (total path length). Distance $\geq |\text{displacement}|$.
- Average velocity = total displacement / total time. Average speed = total distance / total time. Instantaneous velocity = dx/dt .
- For motion with constant acceleration, the three standard equations relate u , v , a , t , and s (displacement).
- On a velocity-time graph: slope = acceleration, area under the graph = displacement.
- For projectile motion (launched at angle θ with speed u): horizontal and vertical motions are independent. Horizontal velocity stays constant; vertical motion is uniformly accelerated under gravity.
- Relative velocity of A with respect to B: $v(AB) = v(A) - v(B)$, added/subtracted as vectors.



Velocity-time graph: the area under the curve gives displacement.

2. Formula Sheet

First equation of motion	$v = u + at$
Second equation of motion	$s = ut + \frac{1}{2} a t^2$
Third equation of motion	$v^2 = u^2 + 2as$
Displacement in nth second	$s_n = u + \frac{a}{2}(2n - 1)$
Time of flight (projectile)	$T = \frac{2u \sin(\theta)}{g}$
Max height (projectile)	$H = \frac{u^2 \sin^2(\theta)}{2g}$
Range (projectile)	$R = \frac{u^2 \sin(2\theta)}{g}$
Max range angle	$\theta = 45^\circ$

3. Practice Problems (22 questions)

1. A car accelerates uniformly from rest to 20 m/s in 10 s. Find its acceleration and the distance covered.
2. A ball is thrown vertically upward with a speed of 30 m/s. Find the maximum height reached and the total time of flight ($g = 10 \text{ m/s}^2$).
3. A body moving with a velocity of 10 m/s is retarded uniformly at 2 m/s^2 . Find the time it takes to stop and the distance travelled before stopping.
4. A stone is dropped from a tower 80 m high. Find the time it takes to reach the ground and its speed on hitting the ground ($g = 10 \text{ m/s}^2$).
5. A projectile is launched with speed 20 m/s at 30 degrees above the horizontal. Find its time of flight, maximum height, and range.
6. A car travels 100 km at 50 km/h and the next 100 km at 25 km/h. Find its average speed for the whole journey.
7. Two trains 100 km apart move towards each other at 40 km/h and 60 km/h. After how much time do they meet?
8. A particle moves with velocity $v = 3t^2 - 2t$ (in m/s). Find its acceleration at $t = 2 \text{ s}$.
9. A body starts from rest and moves with uniform acceleration. Find the ratio of distances covered in the 1st, 2nd, and 3rd seconds.
10. A boat can travel at 10 km/h in still water. It crosses a river 2 km wide flowing at 4 km/h. Find the time taken to cross the river travelling straight across.
11. A ball thrown horizontally from a height of 20 m with speed 15 m/s. Find the time to hit the ground and the horizontal distance travelled.
12. A car moving at 72 km/h is brought to rest by brakes in 5 s. Find the retardation and the distance travelled before stopping.
13. For a body under gravity, sketch the shape of (a) the velocity-time graph and (b) the position-time graph during free fall.

14. A particle covers half of its total distance with speed v_1 and the other half with speed v_2 . Find its average speed.
15. The position of a particle is given by $x = 5t^2 + 3t + 2$ (in meters). Find its velocity and acceleration as functions of time.
16. Two projectiles are fired with the same speed at angles 30 degrees and 60 degrees. Compare their ranges.
17. A lift is moving up with acceleration 2 m/s^2 . A ball is dropped inside the lift from a height of 1.5 m above the floor. Find the time it takes to reach the floor.
18. A cyclist moves in a circle of radius 100 m with constant speed 20 m/s. Find the magnitude of his acceleration.
19. A body is thrown vertically upward. Show that the time of ascent equals the time of descent (neglecting air resistance).
20. A car increases its speed from 36 km/h to 72 km/h in 10 s while travelling on a straight road. Find its acceleration.
21. For a projectile, at what point in its trajectory is the speed minimum, and what is its value?
22. A particle moves along a straight line such that its displacement $s = t^3 - 6t^2 + 9t$. Find the time(s) at which its velocity is zero.

Answer Key

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|---|---|
| 1. $a = 2 \text{ m/s}^2$, $s = 100 \text{ m}$ | 2. $H = 45 \text{ m}$, $T = 6 \text{ s}$ |
| 3. $t = 5 \text{ s}$, $s = 25 \text{ m}$ | 4. $t = 4 \text{ s}$, $v = 40 \text{ m/s}$ |
| 5. $T = 2.04 \text{ s}$, $H = 5.1 \text{ m}$, $R = 35.3 \text{ m}$ | 6. Average speed = 33.3 km/h |
| 7. $t = 1 \text{ h}$ | 8. $a = 10 \text{ m/s}^2$ |
| 9. Ratio 1 : 3 : 5 | 10. $t = 0.2 \text{ h} = 12 \text{ min}$ |
| 11. $t = 2 \text{ s}$, horizontal distance = 30 m | 12. retardation = 4 m/s^2 , $s = 50 \text{ m}$ |
| 13. Conceptual - $v-t$ is a straight line (slope $-g$); $x-t$ is a downward parabola | 14. Average speed = $2 v_1 v_2 / (v_1 + v_2)$ |
| 15. $v = 10t + 3$, $a = 10 \text{ m/s}^2$ (constant) | 16. Equal ranges ($\sin(2 \times 30) = \sin(2 \times 60) = \sin 60$ degrees) |
| 17. $t \approx 0.55 \text{ s}$ | 18. $a = 4 \text{ m/s}^2$ (centripetal) |
| 19. Conceptual - derive using $v = u - gt$ and symmetry | 20. $a = 1 \text{ m/s}^2$ |
| 21. At the highest point, speed = $u \cos(\theta)$ (minimum, horizontal component only) | 22. $t = 1 \text{ s}$ and $t = 3 \text{ s}$ |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.